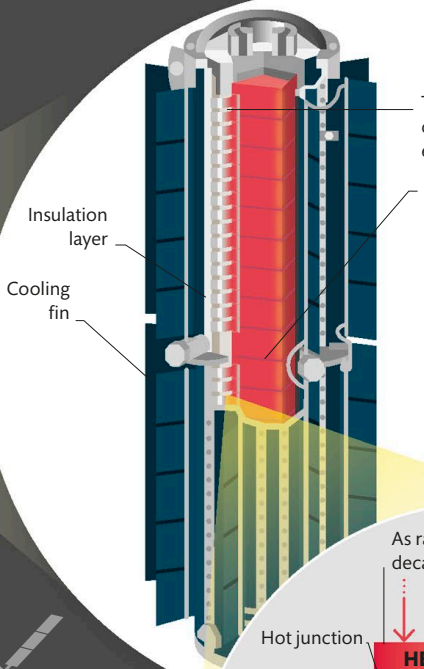
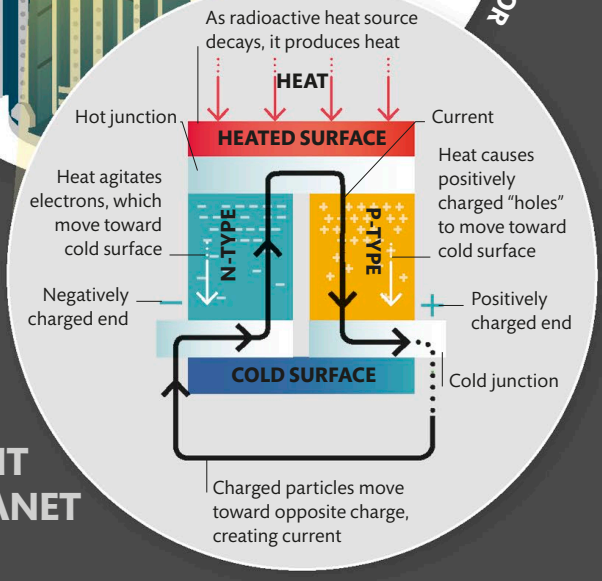
RADIOISOTOPE THERMOELECTRIC GENERATOR



Some nuclear-powered space probes produce electricity by the Seebeck effect. Heat from a radioactive source, such as plutonium, is converted directly into electricity at junctions between two doped semiconductors (see p.160).





IN 1962, **MARINER 2** BECAME THE **FIRST SPACE PROBE** TO **VISIT ANOTHER PLANET**

SPACECRAFT PROPULSION SYSTEMS

Chemical rockets
Rockets burning chemical propellants (see p.61) provide the huge thrust needed to launch probes and make directional corrections and changes in orbits. Gas thrusters are fired to make smaller positional changes.

Ion drive
An ion drive, also known as an ion thruster, uses electricity to accelerate small amounts of electrically charged particles (called ions) into space, generating thrust. Ion drives require fuel to generate electricity.

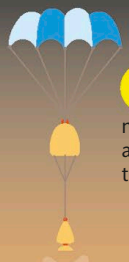
Photon sail
A photon sail, or solar sail, requires no fuel. It uses the radiation pressure of sunlight acting on a giant, mirrorlike sail to propel a spacecraft. The photons in sunlight bounce off the sail, pushing it in the opposite direction.

Landing a probe

Landers use a variety of methods to land on a planet or other object. Typically, parachutes slow the craft as it descends through the atmosphere. Retrorockets slow the descent even more, and then inflated bags may also cushion the landing.



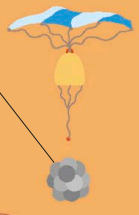
1 Enters atmosphere
After entering the atmosphere, a small pilot chute opens, followed by the main parachute, to slow the lander.



2 Radar
A radar altimeter measures the craft's altitude and triggers the events that follow.



3 Airbags inflate
The heat shield falls away, and large airbags are inflated all around the lander.



4 Landing
Retrorockets fire, and cables holding the lander are cut, letting the craft drop to the surface.



5 On the surface
The lander touches down and bounces across the surface. When it comes to rest, the airbags deflate, and the lander turns itself upright. From entry to landing takes just a few minutes.